

EXPERIMENT 24

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Aim

To write a menu-driven C program using a **switch** statement and nested loops to display five increasing patterns. Each pattern is implemented in a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display the menu containing five pattern choices.
3. Read the user's choice.
4. Use a switch statement to call the corresponding parameterless function.
5. Inside the selected function, input the number of rows n and use nested loops to print the required pattern.
6. For the alphabet pattern, accept only $1 \leq n \leq 26$.
7. If the menu choice is outside 1-5, display an invalid-choice message.
8. Stop.

C Program

```
#include <stdio.h>

void starPattern(void);
void rowNumberPattern(void);
void naturalNumberPattern(void);
void consecutiveNumberPattern(void);
void alphabetPattern(void);

int main(void) {
    int choice;

    printf("INCREASING PATTERNS MENU\n");
    printf("1. Increasing right-angled triangle\n");
    printf("2. Increasing row-number triangle\n");
    printf("3. Increasing natural-number pattern\n");
    printf("4. Consecutive-number triangle\n");
    printf("5. Increasing alphabet pattern\n");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: starPattern(); break;
        case 2: rowNumberPattern(); break;
        case 3: naturalNumberPattern(); break;
        case 4: consecutiveNumberPattern(); break;
        case 5: alphabetPattern(); break;
        default: printf("Invalid choice!\n");
    }
    return 0;
}

void starPattern(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void rowNumberPattern(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", i);
        printf("\n");
    }
}
```

```

void naturalNumberPattern(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}

void consecutiveNumberPattern(void) {
    int n, i, j, num = 1;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= i; j++) printf("%d ", num++);
        printf("\n");
    }
}

void alphabetPattern(void) {
    int n, i, j;
    printf("Enter number of rows (1-26): ");
    scanf("%d", &n);
    if (n < 1 || n > 26) {
        printf("Rows must be between 1 and 26.\n");
        return;
    }
    for (i = 1; i <= n; i++) {
        for (j = 0; j < i; j++) printf("%c ", 'A' + j);
        printf("\n");
    }
}

```

Test-Case Table

Test	Choice	n	Expected result
1	1	4	* / ** / *** / ****
2	2	4	1 / 2 2 / 3 3 3 / 4 4 4 4
3	3	4	1 / 1 2 / 1 2 3 / 1 2 3 4
4	4	4	1 / 2 3 / 4 5 6 / 7 8 9 10
5	5	4	A / A B / A B C / A B C D
6	5	27	Rows must be between 1 and 26.
7	9	-	Invalid choice!

Sample Execution (n = 4)

Choice 1 - Increasing right-angled triangle

```

*
* *
* * *
* * * *

```

Choice 2 - Increasing row-number triangle

```

1
2 2
3 3 3
4 4 4 4

```

Choice 3 - Increasing natural-number pattern

```
1
1 2
1 2 3
1 2 3 4
```

Choice 4 - Consecutive-number triangle

```
1
2 3
4 5 6
7 8 9 10
```

Choice 5 - Increasing alphabet pattern

```
A
A B
A B C
A B C D
```

Result

The program successfully uses a switch statement to select one of five increasing patterns. Each pattern is generated by a separate parameterless void function using nested loops, and the prepared test cases cover all menu options plus invalid inputs.